



1.4

Polynomial Functions and Rates of Change

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: For equally spaced inputs, a degree- n polynomial has constant n th differences.

2. Write one precise sentence explaining this principle: A polynomial rate of change need not be monotonic even when the polynomial is increasing overall.

3. Write one precise sentence explaining this principle: The leading term dominates global behavior, while all terms influence local rates.

4. Write one precise sentence explaining this principle: Secant slopes provide interval-based evidence, not instantaneous conclusions.

Core

5. Compute $P(x + 1) - P(x)$ for $P(x) = x^3 - 2x$.
6. A cubic data table has constant third differences 18 for unit steps. Find the leading coefficient.
7. Why can two cubic models share end behavior but differ strongly on a measured interval?
8. State what the formula $\Delta_h P(x) = P(x + h) - P(x)$ tells you and name any required restriction.
9. State what the formula $\deg(\Delta_h P) = \deg(P) - 1$ ($h \neq 0$) tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Polynomial Functions and Rates of Change. State its domain and justify one key feature.

Medium

11. For $P(x) = x^3 - 2x$, compute $P(x + 1) - P(x)$ and state its degree.
12. For $P(x) = 2x^3 + x^2 - 1$, compute $P(x + 1) - P(x)$ and state its degree.
13. For $P(x) = -x^4 + 3x^2$, compute $P(x + 1) - P(x)$ and state its degree.

14. For $P(x) = 3x^4 - 2x + 5$, compute $P(x+1) - P(x)$ and state its degree.

15. For $P(x) = x^5 - x$, compute $P(x+1) - P(x)$ and state its degree.

Hard

16. A quartic table with unit steps has constant fourth difference 96. Find the leading coefficient.

17. Can a degree-5 polynomial have constant third differences? Explain.

18. For $P(x) = x^4$, compare AROC on $[-2, -1]$ and $[1, 2]$.

19. Find k so $P(x) = x^3 + kx^2$ has AROC 0 on $[-1, 2]$.

Difficult

20. Prove that applying a finite-difference operator to a degree- n polynomial reduces its degree by one.

21. A polynomial has values 1, 4, 15, 40, 85 at $x = 0, 1, 2, 3, 4$. Determine degree evidence and leading coefficient.

22. Construct a quartic with positive leading coefficient and exactly three turning points.

Challenge / HOT

23. If $P(x + 1) - P(x) = 6x^2 + 6x + 2$, find one possible polynomial P .

24. Explain why knowing all interval rates $P(n + 1) - P(n)$ for integer n determines a polynomial P up to one additive constant.

AP-Style Multiple Choice

25. A degree-4 polynomial has constant
(A) first differences
(B) second differences
(C) third differences
(D) fourth differences
26. The degree of $P(x+1) - P(x)$ for cubic P is generally
(A) 0
(B) 1
(C) 2
(D) 3
27. For unit steps, constant third difference 12 implies cubic leading coefficient
(A) 2
(B) 3
(C) 6
(D) 12
28. Which statement is true?
(A) All polynomial rates are constant
(B) Finite differences require equal input steps
(C) End behavior determines all rates
(D) A quartic cannot decrease
29. AROC of x^3 on $[-1, 1]$ is
(A) -1
(B) 0
(C) 1
(D) 2
30. A polynomial with degree n can have at most how many turning points?
(A) n
(B) $n - 1$
(C) $n + 1$
(D) unlimited

AP-Style Free Response

31. A data table at consecutive integers gives 2, 5, 18, 47, 98. (a) Build a finite-difference table. (b) Determine the minimum plausible polynomial degree. (c) Find the leading coefficient. (d) Find the next value.

32. Let $P(x) = x^4 - 4x^2$. (a) Find AROC on $[-2, 0]$, $[0, 2]$, and $[-2, 2]$. (b) Explain the results using symmetry. (c) Give an interval with positive AROC.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.